

Theory Assignment 2- 2025-2026
Introduction to Databases (CSE 3151)
Programme: B.Tech. (CSE & CSIT) Semester: 6th
Submission Date: 4 May 2026

Course Learning Outcome	*Taxonomy Level
Apply relational algebra and relational calculus to express queries on relational schemas.	L4

This assignment is designed to give you practice with the concepts of

- Expressing queries using formal relational query languages

1. Consider the following relational schemas

- a. **Employee** (ename, ecity, state)
- b. **Works** (ename, company name, salary)
- c. **Company** (company name, city)
- d. **Manages**(ename, managername)

Write the expressions to solve the following queries using **Relational Algebra, Tuple Relational Calculus, and Domain relational Calculus**.

- a. Write the expressions to find the name of the employees getting salary more than 60,000 and working in TCS and WIPRO.
- b. Write the expressions to find the salary and name of the employees working in any company placed in DELHI.
- c. Write the expression to find the salary of the employees living in city MUMBAI and working under the manager JOHN.

2. Consider the following relational schemas

- **Student**(SID, Name, City, Age)
- **Course**(CID, CName, Dept, Credits)
- **Enroll**(SID, CID, Marks)
- **Teacher**(TID, TName, Dept, Salary)
- **Teaches**(TID, CID)

Write the expressions to solve the following queries using **Relational Algebra**.

- a) Find out the all ITD teachers with their names from CSE department who have the salary is less than 80000.
- b) Find out the names of all students with their marks and credits they are enrolled in any courses.
- c) Find out the courses taught by more than one distinct teacher.

3. Consider the following three relations in a relational database.

- **Employee**(Eld, Name),
- **Brand**(Bld, BName),
- **Own**(Eld, Bld)

Write the relational algebra expression that will return the set of Elds who own all the brands. Write the equivalent tuple relational calculus and domain relational calculus query.

-----**End**-----